Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: A & M de Sousa Dairy

Physical address of dairy:

Number and Street	City	County	Zip Code
3030 S Central AVE	Turlock	Stanislaus	95380

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1920

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

0058-0017-0005-0000 0058-0017-0006-0000

B. OPERATORS

Sousa, Armelim

Operator name: Sousa, Armelim

Telephone no.: (209) 632-5845 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
3030 S Central AVE	Turlock	CA	95380

This operator is responsible for paying permit fees.

C. OWNERS

Sousa, Armelim

Legal owner name: Sousa, Armelim

Telephone no.: (209) 632-5845 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
3030 S Central AVE	Turlock	CA	95380

This owner is responsible for paying permit fees.

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AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	275	210	430	280	30
Number under roof	1,212	0	0	0	0	0
Maximum number	1,212	275	210	430	280	30
Average number	1,212	275	210	430	280	30
Avg live weight (lbs)	1,400	1,450	1,000	700		

Predominant milk cow breed: Jersey-Holstein Cross

Average milk production: 81 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 44,023.16 *tons per reporting period*

Total nitrogen from manure: 553,702.60 *lbs per reporting period*

After ammonia losses (30% loss applied): 387,591.82 *lbs per reporting period*

Total phosphorus from manure: 90,599.63 *lbs per reporting period*

Total potassium from manure: 240,868.31 *lbs per reporting period*

Total salt from manure: 633,906.45 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 44,208,026 *gallons*

Total nitrogen generated: 142,972.02 *lbs*

Total phosphorus generated: 15,804.76 *lbs*

Total potassium generated: 163,760.03 *lbs*

Total salt generated: 1,243,903.67 *lbs*

	44,208,026	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	44,208,026	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Ag Well 3	Ground water
San Joaquin River	Surface water
Turlock Irrigation District	Surface water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

Date	Material type / Description	Quantity	Reporting basis	Moisture (%)	N (%)	P (%)	K (%)	Salt (%)
06/30/2023	Liquid commercial fertilizer *Fertilizer*	4,587.00 *gal*			32.000000	0.000000	0.000000	0.000000

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Commercial fertilizer / Other	12,249.12	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Total imports for all materials	12,249.12	0.00	0.00	0.00

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

No liquid nutrient exports entered.

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
A1-Dairy Field	121	121	2	process wastewater	X058-X017-X003-XXXX X058-X017-X004-XXXX X058-X017-X005-XXXX X058-X017-X006-XXXX
A3-4 (Silva, Lima)	92	92	2	process wastewater	X058-X018-X010-XXXX X058-X018-X013-XXXX
A5 (Linwood, Gomes, Heifer)	74	74	1	process wastewater	X058-X009-X004-XXXX X058-X011-X003-XXXX X058-X011-X032-XXXX
A6 (Brown)	60	60	2	manure	X058-X027-X002-XXXX X058-X027-X003-XXXX X058-X027-X004-XXXX X058-X027-X007-XXXX X058-X029-X005-XXXX
A8 (Joe Borba-Moffett)	92	92	2	manure	X057-X015-X001-XXXX X058-X028-X003-XXXX
B1-2 (August)	125	125	2	manure	X057-X024-X001-XXXX
B3-4 (King)	160	160	1	manure	X057-X025-X013-XXXX
Borba Central	38	38	2	process wastewater	X058-X028-X022-XXXX
Borba-Blaker	38	38	2	process wastewater	X058-X019-X004-XXXX
R1-2 (River)	100	100	1	manure	X057-X025-X009-XXXX
Totals for areas that were used for application	900	900	17		
Totals for areas that were not used for application					
Land application area totals	900	900	17		

B. CROPS AND HARVESTS

A1-Dairy Field

Field name: A1-Dairy Field

A1-Dairy Field

11/11/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 121 Plant date: 11/11/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/27/2023	1,147.00 *ton*	Dry-weight		66.6	32,300.00	3,080.00	29,700.00		10.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	1,600.00
Total actual harvest content	9.48	204.53	19.50	188.07	658.55

06/10/2023: Corn, silage

Crop: Corn, silage Acres planted: 121 Plant date: 06/10/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/19/2023	3,588.00 *ton*	Dry-weight		70.8	12,000.00	1,970.00	15,900.00		6.57

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	1,500.00
Total actual harvest content	29.65	207.81	34.12	275.34	1,137.75

A3-4 (Silva, Lima)

Field name: A3-4 (Silva, Lima)

11/13/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 92 Plant date: 11/13/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/21/2023	1,256.00 *ton*	Dry-weight		77.6	32,500.00	5,000.00	59,000.00		12.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	1,600.00
Total actual harvest content	13.65	198.78	30.58	360.85	764.52

A3-4 (Silva, Lima)

05/14/2023: Corn, silage

Crop: Corn, silage Acres planted: 92 Plant date: 05/14/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/11/2023	2,667.00 *ton*	Dry-weight		67.9	12,100.00	1,950.00	14,200.00		5.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	1,500.00
Total actual harvest content	28.99	225.19	36.29	264.28	986.38

A5 (Linwood, Gomes, Heifer)

Field name: A5 (Linwood, Gomes, Heifer)

08/05/2022: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 74 Plant date: 08/05/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
07/20/2023	590.00 *ton*	Dry-weight		8.3	33,900.00	2,000.00	15,300.00		5.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	480.00	43.20	336.00	800.00
Total actual harvest content	7.97	495.70	29.24	223.72	731.12

A6 (Brown)

Field name: A6 (Brown)

A6 (Brown)

11/14/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 60 Plant date: 11/14/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/17/2023	556.00 *ton*	Dry-weight		68.4	34,700.00	4,400.00	38,000.00		10.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	1,600.00
Total actual harvest content	9.27	203.22	25.77	222.55	591.51

05/17/2023: Corn, silage

Crop: Corn, silage Acres planted: 60 Plant date: 05/17/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/19/2023	1,773.00 *ton*	Dry-weight		64.6	10,300.00	1,470.00	10,800.00		5.68

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	1,500.00
Total actual harvest content	29.55	215.49	30.75	225.95	1,188.34

A8 (Joe Borba-Moffett)

Field name: A8 (Joe Borba-Moffett)

11/17/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 92 Plant date: 11/17/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/10/2023	1,408.00 *ton*	Dry-weight		71.8	21,200.00	3,560.00	29,200.00		10.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	1,600.00
Total actual harvest content	15.30	182.99	30.73	252.04	923.59

A8 (Joe Borba-Moffett)

06/14/2023: Corn, silage

Crop: Corn, silage Acres planted: 92 Plant date: 06/14/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/02/2023	2,771.00 *ton*	Dry-weight		74.8	9,000.00	2,230.00	12,700.00		4.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	1,500.00
Total actual harvest content	30.12	136.62	33.85	192.79	607.21

B1-2 (August)

Field name: B1-2 (August)

10/30/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 125 Plant date: 10/30/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/22/2023	1,981.00 *ton*	Dry-weight		55.6	7,550.00	2,230.00	10,600.00		9.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	1,600.00
Total actual harvest content	15.85	106.25	31.38	149.17	1,351.01

06/08/2023: Corn, silage

Crop: Corn, silage Acres planted: 125 Plant date: 06/08/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
11/21/2023	2,876.00 *ton*	Dry-weight		70.5	10,700.00	1,460.00	14,800.00		5.62

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	165.00	1,250.00
Total actual harvest content	23.01	145.25	19.82	200.91	762.90

B3-4 (King)

Field name: B3-4 (King)

07/20/2023: Sudangrass, silage

Crop: Sudangrass, silage Acres planted: 160 Plant date: 07/20/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
11/20/2023	1,438.00 *ton*	Dry-weight		66.6	32,300.00	3,080.00	29,700.00		10.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	20.00	220.00	34.00	240.00	2,000.00
Total actual harvest content	8.99	193.92	18.49	178.31	624.38

Borba Central

Field name: Borba Central

10/25/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 38 Plant date: 10/25/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/12/2023	598.00 *ton*	Dry-weight		63.8	15,400.00	2,970.00	27,900.00		11.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	1,600.00
Total actual harvest content	15.74	175.46	33.84	317.88	1,264.68

05/20/2023: Corn, silage

Crop: Corn, silage Acres planted: 38 Plant date: 05/20/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/10/2023	1,151.00 *ton*	Dry-weight		71.3	11,300.00	2,020.00	14,600.00		5.46

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	1,500.00
Total actual harvest content	30.29	196.46	35.12	253.84	949.28

Borba-Blaker

Field name: Borba-Blaker

10/22/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 38 Plant date: 10/22/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/24/2023	557.00 *ton*	Dry-weight		74.2	18,300.00	2,820.00	28,000.00		10.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	1,600.00
Total actual harvest content	14.66	138.41	21.33	211.78	801.73

05/16/2023: Corn, silage

Crop: Corn, silage Acres planted: 38 Plant date: 05/16/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
11/22/2023	981.00 *ton*	Dry-weight		56.6	8,230.00	1,470.00	5,750.00		4.17

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	1,500.00
Total actual harvest content	25.82	184.42	32.94	128.85	934.42

R1-2 (River)

Field name: R1-2 (River)

07/17/2023: Sudangrass, silage

Crop: Sudangrass, silage Acres planted: 100 Plant date: 07/17/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
11/20/2023	1,723.00 *ton*	Dry-weight		71.8	21,200.00	3,560.00	29,200.00		10.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	20.00	220.00	34.00	240.00	2,000.00
Total actual harvest content	17.23	206.02	34.60	283.76	1,039.80

NUTRIENT BUDGET

A. LAND APPLICATIONS

A1-Dairy Field - 11/11/2022: Oats, silage-soft dough

Field name: A1-Dairy Field

Crop: Oats, silage-soft dough — Plant date: 11/11/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/29/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	61.87	3.09	68.28	533.24	1,347,000.00 *gal*
Turlock Irrigation District	Surface water	1.65	0.00	0.00	303.92	14,989,000.00 *gal*
Application event totals		63.52	3.09	68.28	837.16	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	137.20	6.86	151.41	1,182.46	2,987,000.00 *gal*
Turlock Irrigation District	Surface water	1.66	0.00	0.00	304.41	15,013,000.00 *gal*
Application event totals		138.86	6.86	151.41	1,486.87	

A1-Dairy Field - 06/10/2023: Corn, silage

Field name: A1-Dairy Field

Crop: Corn, silage — Plant date: 06/10/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	62.45	9.56	82.79	528.12	2,587,000.00 *gal*
Turlock Irrigation District	Surface water	1.78	0.00	0.00	327.40	16,147,000.00 *gal*
Application event totals		64.23	9.56	82.79	855.52	

A1-Dairy Field - 06/10/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Turlock Irrigation District	Surface water	1.82	0.00	0.00	333.69	16,457,000.00 *gal*
Application event totals		1.82	0.00	0.00	333.69	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	52.11	7.98	69.09	440.74	2,159,000.00 *gal*
Turlock Irrigation District	Surface water	1.77	0.00	0.00	324.89	16,023,000.00 *gal*
Application event totals		53.88	7.98	69.09	765.63	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	57.16	8.75	75.78	483.41	2,368,000.00 *gal*
Turlock Irrigation District	Surface water	1.77	0.00	0.00	324.36	15,997,000.00 *gal*
Application event totals		58.93	8.75	75.78	807.77	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	60.27	9.23	79.91	509.74	2,497,000.00 *gal*
Turlock Irrigation District	Surface water	1.77	0.00	0.00	324.42	16,000,000.00 *gal*
Application event totals		62.04	9.23	79.91	834.16	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	35.31	5.41	46.82	298.66	1,463,000.00 *gal*
Turlock Irrigation District	Surface water	1.77	0.00	0.00	324.42	16,000,000.00 *gal*
Application event totals		37.08	5.41	46.82	623.08	

A1-Dairy Field - 06/10/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Turlock Irrigation District	Surface water	1.77	0.00	0.00	324.42	16,000,000.00 *gal*
Application event totals		1.77	0.00	0.00	324.42	

A3-4 (Silva, Lima) - 11/13/2022: Oats, silage-soft dough

Field name: A3-4 (Silva, Lima)

Crop: Oats, silage-soft dough Plant date: 11/13/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/28/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	92.19	4.61	101.74	794.52	1,526,000.00 *gal*
Turlock Irrigation District	Surface water	2.46	0.00	0.00	451.40	16,927,000.00 *gal*
Application event totals		94.64	4.61	101.74	1,245.92	

04/12/2023	Subsurface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	90.13	4.51	99.47	776.82	1,492,000.00 *gal*
Turlock Irrigation District	Surface water	2.21	0.00	0.00	406.15	15,230,000.00 *gal*
Application event totals		92.34	4.51	99.47	1,182.97	

A3-4 (Silva, Lima) - 05/14/2023: Corn, silage

Field name: A3-4 (Silva, Lima)

Crop: Corn, silage Plant date: 05/14/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

A3-4 (Silva, Lima) - 05/14/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	58.92	9.02	78.12	498.32	1,856,000.00 *gal*
Turlock Irrigation District	Surface water	2.47	0.00	0.00	453.63	17,010,268.00 *gal*
Application event totals		61.39	9.02	78.12	951.94	

06/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Turlock Irrigation District	Surface water	2.32	0.00	0.00	426.68	16,000,000.00 *gal*
Application event totals		2.32	0.00	0.00	426.68	

07/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	76.92	11.78	101.98	650.55	2,423,000.00 *gal*
Turlock Irrigation District	Surface water	2.18	0.00	0.00	400.02	15,000,000.00 *gal*
Application event totals		79.10	11.78	101.98	1,050.57	

07/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	63.08	9.66	83.63	533.49	1,987,000.00 *gal*
Turlock Irrigation District	Surface water	2.10	0.00	0.00	386.68	14,500,000.00 *gal*
Application event totals		65.19	9.66	83.63	920.17	

07/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	32.91	0.89	6.63	340.76	1,423,000.00 *gal*
Turlock Irrigation District	Surface water	1.81	0.00	0.00	333.35	12,500,000.00 *gal*
Application event totals		34.73	0.89	6.63	674.11	

A3-4 (Silva, Lima) - 05/14/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	59.21	9.07	78.49	500.74	1,865,000.00 *gal*
Turlock Irrigation District	Surface water	1.67	0.00	0.00	306.68	11,500,000.00 *gal*
Application event totals		60.88	9.07	78.49	807.42	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Turlock Irrigation District	Surface water	2.18	0.00	0.00	400.02	15,000,000.00 *gal*
Application event totals		2.18	0.00	0.00	400.02	

A5 (Linwood, Gomes, Heifer) - 08/05/2022: Alfalfa, hay

Field name: A5 (Linwood, Gomes, Heifer)

Crop: Alfalfa, hay

Plant date: 08/05/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	84.27	12.90	111.71	712.66	2,135,000.00 *gal*
Turlock Irrigation District	Surface water	2.07	0.00	0.00	381.28	11,500,000.00 *gal*
Application event totals		86.34	12.90	111.71	1,093.94	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Turlock Irrigation District	Surface water	2.07	0.00	0.00	381.28	11,500,000.00 *gal*
Application event totals		2.07	0.00	0.00	381.28	

A5 (Linwood, Gomes, Heifer) - 08/05/2022: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Turlock Irrigation District	Surface water	2.07	0.00	0.00	381.28	11,500,000.00 *gal*
Application event totals		2.07	0.00	0.00	381.28	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Turlock Irrigation District	Surface water	2.07	0.00	0.00	381.28	11,500,000.00 *gal*
Application event totals		2.07	0.00	0.00	381.28	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Turlock Irrigation District	Surface water	2.07	0.00	0.00	381.28	11,500,000.00 *gal*
Application event totals		2.07	0.00	0.00	381.28	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	57.14	1.54	11.52	591.56	1,987,000.00 *gal*
Turlock Irrigation District	Surface water	2.07	0.00	0.00	381.28	11,500,000.00 *gal*
Application event totals		59.21	1.54	11.52	972.83	

A6 (Brown) - 11/14/2022: Oats, silage-soft dough

Field name: A6 (Brown)

Crop: Oats, silage-soft dough

Plant date: 11/14/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/22/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure	Separator solids	190.18	47.82	45.86	399.14	1,283.00 *ton*
Application event totals		190.18	47.82	45.86	399.14	

A6 (Brown) - 11/14/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/25/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Turlock Irrigation District	Surface water	2.01	0.00	0.00	368.51	9,012,000.00 *gal*
Application event totals		2.01	0.00	0.00	368.51	

04/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Turlock Irrigation District	Surface water	2.00	0.00	0.00	367.89	8,997,000.00 *gal*
Application event totals		2.00	0.00	0.00	367.89	

A6 (Brown) - 05/17/2023: Corn, silage

Field name: A6 (Brown)

Crop: Corn, silage

Plant date: 05/17/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Turlock Irrigation District	Surface water	2.05	0.00	0.00	377.46	9,231,000.00 *gal*
Application event totals		2.05	0.00	0.00	377.46	

05/08/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure	Separator solids	172.10	43.27	41.50	361.19	1,161.00 *ton*
Application event totals		172.10	43.27	41.50	361.19	

07/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Fertilizer	Liquid commercial fertilizer	50.00	0.00	0.00	0.00	
Turlock Irrigation District	Surface water	2.00	0.00	0.00	366.95	8,974,000.00 *gal*
Application event totals		52.00	0.00	0.00	366.95	

A6 (Brown) - 05/17/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Turlock Irrigation District	Surface water	1.78	0.00	0.00	327.12	8,000,000.00 *gal*
Application event totals		1.78	0.00	0.00	327.12	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Fertilizer	Liquid commercial fertilizer	50.00	0.00	0.00	0.00	
Turlock Irrigation District	Surface water	1.78	0.00	0.00	327.12	8,000,000.00 *gal*
Application event totals		51.78	0.00	0.00	327.12	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Turlock Irrigation District	Surface water	1.80	0.00	0.00	331.58	8,109,000.00 *gal*
Application event totals		1.80	0.00	0.00	331.58	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Turlock Irrigation District	Surface water	1.78	0.00	0.00	327.12	8,000,000.00 *gal*
Application event totals		1.78	0.00	0.00	327.12	

A8 (Joe Borba-Moffett) - 11/17/2022: Oats, silage-soft dough

Field name: A8 (Joe Borba-Moffett)

Crop: Oats, silage-soft dough

Plant date: 11/17/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/28/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure	Corral solids	195.49	192.10	490.53	17,011.47	1,453.00 *ton*
Application event totals		195.49	192.10	490.53	17,011.47	

A8 (Joe Borba-Moffett) - 11/17/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/01/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Turlock Irrigation District	Surface water	1.91	0.00	0.00	350.87	13,157,000.00 *gal*
Application event totals		1.91	0.00	0.00	350.87	

04/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Turlock Irrigation District	Surface water	1.79	0.00	0.00	329.53	12,356,951.00 *gal*
Application event totals		1.79	0.00	0.00	329.53	

A8 (Joe Borba-Moffett) - 06/14/2023: Corn, silage

Field name: A8 (Joe Borba-Moffett)

Crop: Corn, silage

Plant date: 06/14/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Turlock Irrigation District	Surface water	2.04	0.00	0.00	373.96	14,023,000.00 *gal*
Application event totals		2.04	0.00	0.00	373.96	

06/05/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure	Separator solids	137.56	34.59	33.17	288.71	1,423.00 *ton*
Application event totals		137.56	34.59	33.17	288.71	

07/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Turlock Irrigation District	Surface water	1.96	0.00	0.00	360.01	13,500,000.00 *gal*
Application event totals		1.96	0.00	0.00	360.01	

A8 (Joe Borba-Moffett) - 06/14/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Turlock Irrigation District	Surface water	1.97	0.00	0.00	361.80	13,567,000.00 *gal*
Application event totals		1.97	0.00	0.00	361.80	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Turlock Irrigation District	Surface water	1.87	0.00	0.00	343.80	12,892,000.00 *gal*
Application event totals		1.87	0.00	0.00	343.80	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Turlock Irrigation District	Surface water	1.87	0.00	0.00	343.80	12,892,000.00 *gal*
Application event totals		1.87	0.00	0.00	343.80	

B1-2 (August) - 10/30/2022: Oats, silage-soft dough

Field name: B1-2 (August)

Crop: Oats, silage-soft dough

Plant date: 10/30/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/15/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well 3	Ground water	0.00	0.00	0.00	1,273.23	18,268,000.00 *gal*
Application event totals		0.00	0.00	0.00	1,273.23	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/11/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure	Separator solids	103.67	26.07	25.00	217.57	1,457.00 *ton*
Application event totals		103.67	26.07	25.00	217.57	

B1-2 (August) - 10/30/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well 3	Ground water	0.00	0.00	0.00	1,185.57	17,010,268.00 *gal*
Application event totals		0.00	0.00	0.00	1,185.57	

B1-2 (August) - 06/08/2023: Corn, silage

Field name: B1-2 (August)

Crop: Corn, silage

Plant date: 06/08/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/13/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure	Separator solids	141.38	35.55	34.09	296.71	1,987.00 *ton*
Application event totals		141.38	35.55	34.09	296.71	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well 3	Ground water	0.00	0.00	0.00	1,254.62	18,001,000.00 *gal*
Application event totals		0.00	0.00	0.00	1,254.62	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Fertilizer	Liquid commercial fertilizer	50.00	0.00	0.00	0.00	
Ag Well 3	Ground water	0.00	0.00	0.00	1,219.71	17,500,000.00 *gal*
Application event totals		50.00	0.00	0.00	1,219.71	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well 3	Ground water	0.00	0.00	0.00	1,185.57	17,010,268.00 *gal*
Application event totals		0.00	0.00	0.00	1,185.57	

B1-2 (August) - 06/08/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well 3	Ground water	0.00	0.00	0.00	1,114.60	15,992,000.00 *gal*
Application event totals		0.00	0.00	0.00	1,114.60	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well 3	Ground water	0.00	0.00	0.00	1,115.16	16,000,000.00 *gal*
Application event totals		0.00	0.00	0.00	1,115.16	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well 3	Ground water	0.00	0.00	0.00	1,131.54	16,235,000.00 *gal*
Application event totals		0.00	0.00	0.00	1,131.54	

B3-4 (King) - 07/20/2023: Sudangrass, silage

Field name: B3-4 (King)

Crop: Sudangrass, silage

Plant date: 07/20/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well 3	Ground water	0.00	0.00	0.00	926.23	17,010,268.00 *gal*
Application event totals		0.00	0.00	0.00	926.23	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/20/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure	Corral solids	179.86	176.75	451.33	15,651.90	2,325.00 *ton*
Application event totals		179.86	176.75	451.33	15,651.90	

B3-4 (King) - 07/20/2023: Sudangrass, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well 3	Ground water	0.00	0.00	0.00	1,013.01	18,604,000.00 *gal*
Application event totals		0.00	0.00	0.00	1,013.01	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well 3	Ground water	0.00	0.00	0.00	979.41	17,987,000.00 *gal*
Application event totals		0.00	0.00	0.00	979.41	

Borba Central - 10/25/2022: Oats, silage-soft dough

Field name: Borba Central

Crop: Oats, silage-soft dough

Plant date: 10/25/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/26/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	121.98	18.68	161.71	1,031.60	1,587,000.00 *gal*
Turlock Irrigation District	Surface water	1.76	0.00	0.00	322.82	5,000,000.00 *gal*
Application event totals		123.74	18.68	161.71	1,354.42	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	88.16	13.50	116.88	745.59	1,147,000.00 *gal*
Turlock Irrigation District	Surface water	1.71	0.00	0.00	314.56	4,872,000.00 *gal*
Application event totals		89.87	13.50	116.88	1,060.14	

Borba Central - 05/20/2023: Corn, silage

Borba Central - 05/20/2023: Corn, silage

Field name: Borba Central

Crop: Corn, silage

Plant date: 05/20/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	51.69	1.39	10.42	535.12	923,000.00 *gal*
Turlock Irrigation District	Surface water	2.15	0.00	0.00	394.42	6,109,000.00 *gal*
Application event totals		53.83	1.39	10.42	929.54	

06/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Turlock Irrigation District	Surface water	1.93	0.00	0.00	355.10	5,500,000.00 *gal*
Application event totals		1.93	0.00	0.00	355.10	

07/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	58.11	8.90	77.03	491.42	756,000.00 *gal*
Turlock Irrigation District	Surface water	1.91	0.00	0.00	350.52	5,429,080.00 *gal*
Application event totals		60.02	8.90	77.03	841.95	

07/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	43.40	1.17	8.75	449.31	775,000.00 *gal*
Turlock Irrigation District	Surface water	1.93	0.00	0.00	355.10	5,500,000.00 *gal*
Application event totals		45.33	1.17	8.75	804.41	

07/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	41.83	1.13	8.43	433.08	747,000.00 *gal*
Turlock Irrigation District	Surface water	1.91	0.00	0.00	350.52	5,429,080.00 *gal*
Application event totals		43.74	1.13	8.43	783.60	

Borba Central - 05/20/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	60.18	9.22	79.79	508.99	783,026.00 *gal*
Turlock Irrigation District	Surface water	1.91	0.00	0.00	350.52	5,429,080.00 *gal*
Application event totals		62.09	9.22	79.79	859.51	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Turlock Irrigation District	Surface water	1.93	0.00	0.00	355.10	5,500,000.00 *gal*
Application event totals		1.93	0.00	0.00	355.10	

Borba-Blaker - 10/22/2022: Oats, silage-soft dough

Field name: Borba-Blaker

Crop: Oats, silage-soft dough

Plant date: 10/22/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/22/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	101.77	15.58	134.91	860.64	1,324,000.00 *gal*
Turlock Irrigation District	Surface water	1.93	0.00	0.00	355.10	5,500,000.00 *gal*
Application event totals		103.70	15.58	134.91	1,215.74	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	63.17	1.70	12.73	653.97	1,128,000.00 *gal*
Turlock Irrigation District	Surface water	1.91	0.00	0.00	350.52	5,429,080.00 *gal*
Application event totals		65.07	1.70	12.73	1,004.49	

Borba-Blaker - 05/16/2023: Corn, silage

Borba-Blaker - 05/16/2023: Corn, silage

Field name: Borba-Blaker

Crop: Corn, silage

Plant date: 05/16/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	91.62	14.03	121.46	774.84	1,192,000.00 *gal*
Turlock Irrigation District	Surface water	2.14	0.00	0.00	393.97	6,102,000.00 *gal*
Application event totals		93.76	14.03	121.46	1,168.81	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Turlock Irrigation District	Surface water	1.93	0.00	0.00	355.10	5,500,000.00 *gal*
Application event totals		1.93	0.00	0.00	355.10	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	65.87	10.09	87.33	557.08	857,000.00 *gal*
Turlock Irrigation District	Surface water	1.91	0.00	0.00	350.52	5,429,080.00 *gal*
Application event totals		67.78	10.09	87.33	907.60	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Turlock Irrigation District	Surface water	1.93	0.00	0.00	355.10	5,500,000.00 *gal*
Application event totals		1.93	0.00	0.00	355.10	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	68.18	10.44	90.38	576.58	887,000.00 *gal*
Turlock Irrigation District	Surface water	1.91	0.00	0.00	350.52	5,429,080.00 *gal*
Application event totals		70.08	10.44	90.38	927.10	

Borba-Blaker - 05/16/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Turlock Irrigation District	Surface water	1.93	0.00	0.00	355.10	5,500,000.00 *gal*
Application event totals		1.93	0.00	0.00	355.10	

08/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Turlock Irrigation District	Surface water	1.91	0.00	0.00	350.52	5,429,080.00 *gal*
Application event totals		1.91	0.00	0.00	350.52	

R1-2 (River) - 07/17/2023: Sudangrass, silage

Field name: R1-2 (River)

Crop: Sudangrass, silage

Plant date: 07/17/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
San Joaquin River	Surface water	0.49	0.00	0.00	242.33	12,356,951.00 *gal*
Application event totals		0.49	0.00	0.00	242.33	

07/20/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure	Corral solids	260.30	255.80	653.17	22,651.83	2,103.00 *ton*
Application event totals		260.30	255.80	653.17	22,651.83	

08/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
San Joaquin River	Surface water	0.49	0.00	0.00	242.33	12,356,951.00 *gal*
Application event totals		0.49	0.00	0.00	242.33	

R1-2 (River) - 07/17/2023: Sudangrass, silage

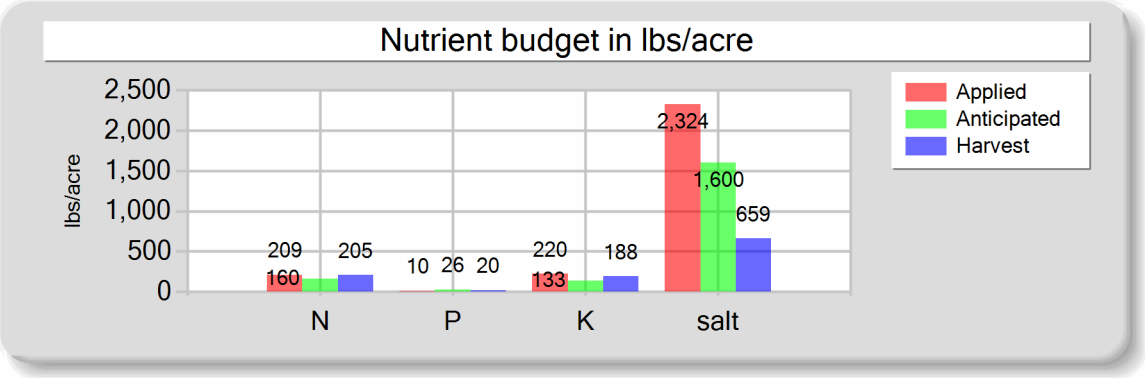
Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
San Joaquin River	Surface water	0.49	0.00	0.00	242.33	12,356,951.00 *gal*
Application event totals		0.49	0.00	0.00	242.33	

B. NUTRIENT BUDGET

A1-Dairy Field - 11/11/2022: Oats, silage-soft dough

Field name: A1-Dairy Field Crop: Oats, silage-soft dough Plant date: 11/11/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	199.07	9.95	219.69	1,715.70
Fresh water	3.31	0.00	0.00	608.33
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	209.38	9.95	219.69	2,324.03
Anticipated crop nutrient removal	160.00	25.60	132.80	1,600.00
Actual crop nutrient removal	204.53	19.50	188.07	658.55
Nutrient balance	4.85	-9.55	31.63	1,665.48
Applied to removed ratio	1.02	0.51	1.17	3.53

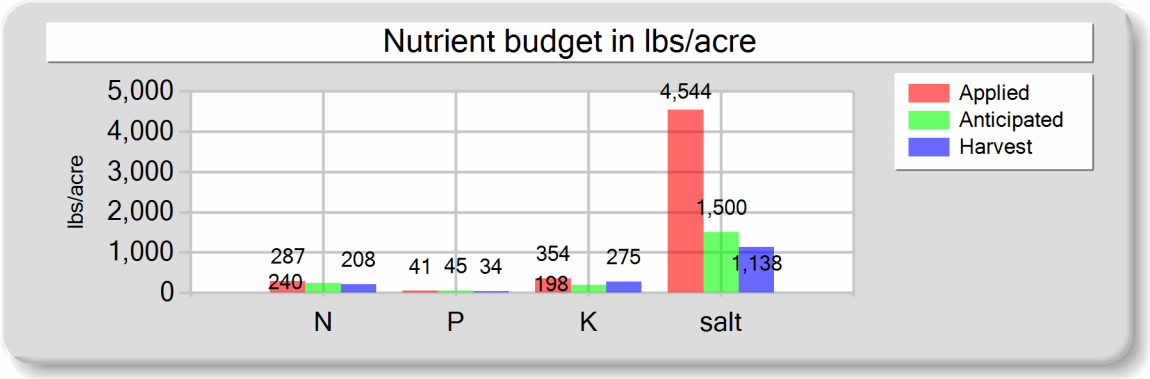
Fresh water applied	
30,002,000.00	*gallons*
1,104.87	*acre-inches*
9.13	*inches/acre*

Process wastewater applied	
4,334,000.00	*gallons*
159.61	*acre-inches*
1.32	*inches/acre*

Total harvests for the crop	
1	*harvests*

A1-Dairy Field - 06/10/2023: Corn, silage

Field name: A1-Dairy Field Crop: Corn, silage Plant date: 06/10/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	267.31	40.94	354.38	2,260.67
Fresh water	12.43	0.00	0.00	2,283.60
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	286.74	40.94	354.38	4,544.27
Anticipated crop nutrient removal	240.00	45.00	198.00	1,500.00
Actual crop nutrient removal	207.81	34.12	275.34	1,137.75
Nutrient balance	78.93	6.82	79.03	3,406.52
Applied to removed ratio	1.38	1.20	1.29	3.99

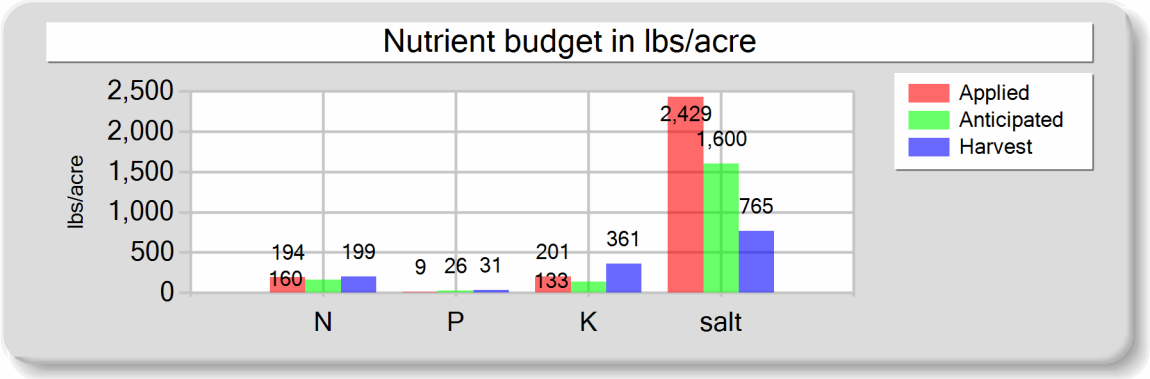
Fresh water applied	
112,624,000.00	*gallons*
4,147.56	*acre-inches*
34.28	*inches/acre*

Process wastewater applied	
11,074,000.00	*gallons*
407.82	*acre-inches*
3.37	*inches/acre*

Total harvests for the crop	
1	*harvests*

A3-4 (Silva, Lima) - 11/13/2022: Oats, silage-soft dough

Field name: A3-4 (Silva, Lima) Crop: Oats, silage-soft dough Plant date: 11/13/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	182.32	9.12	201.21	1,571.34
Fresh water	4.67	0.00	0.00	857.55
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	193.99	9.12	201.21	2,428.89
Anticipated crop nutrient removal	160.00	25.60	132.80	1,600.00
Actual crop nutrient removal	198.78	30.58	360.85	764.52
Nutrient balance	-4.79	-21.46	-159.65	1,664.37
Applied to removed ratio	0.98	0.30	0.56	3.18

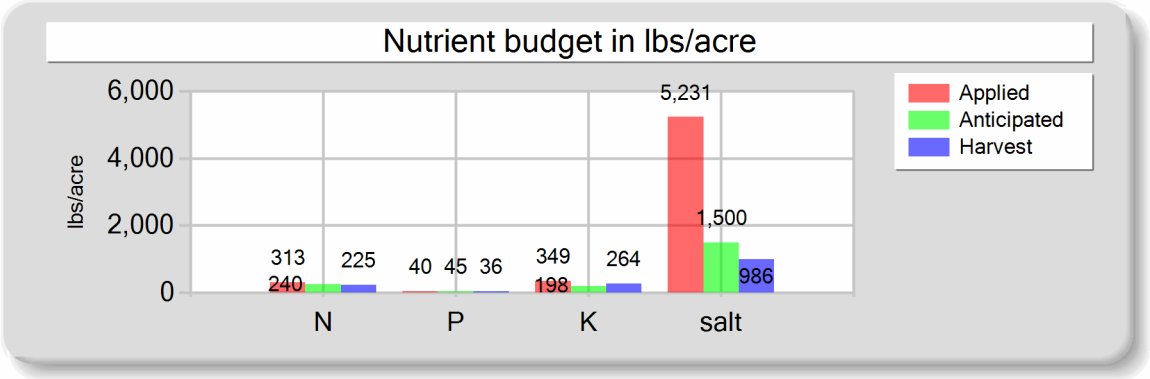
Fresh water applied	
32,157,000.00	*gallons*
1,184.23	*acre-inches*
12.87	*inches/acre*

Process wastewater applied	
3,018,000.00	*gallons*
111.14	*acre-inches*
1.21	*inches/acre*

Total harvests for the crop	
1	*harvests*

A3-4 (Silva, Lima) - 05/14/2023: Corn, silage

Field name: A3-4 (Silva, Lima) Crop: Corn, silage Plant date: 05/14/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	291.05	40.42	348.85	2,523.86
Fresh water	14.73	0.00	0.00	2,707.05
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	312.78	40.42	348.85	5,230.91
Anticipated crop nutrient removal	240.00	45.00	198.00	1,500.00
Actual crop nutrient removal	225.19	36.29	264.28	986.38
Nutrient balance	87.59	4.13	84.57	4,244.52
Applied to removed ratio	1.39	1.11	1.32	5.30

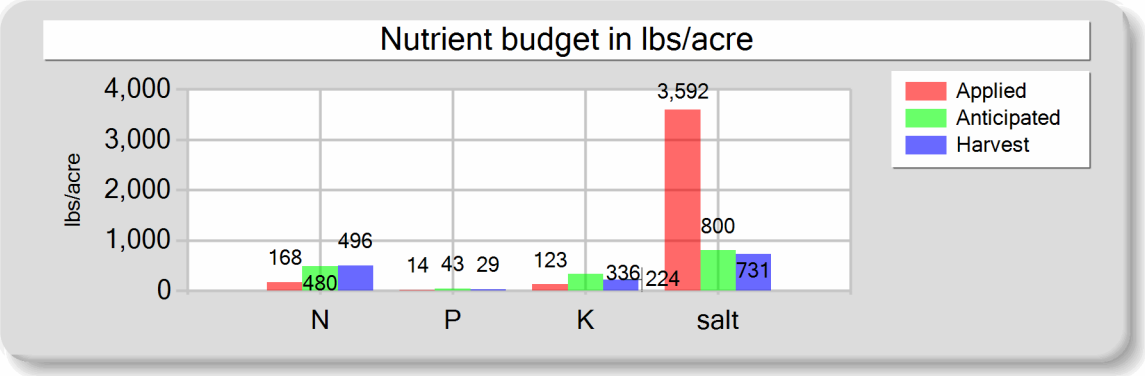
Fresh water applied	
101,510,268.00	*gallons*
3,738.28	*acre-inches*
40.63	*inches/acre*

Process wastewater applied	
9,554,000.00	*gallons*
351.84	*acre-inches*
3.82	*inches/acre*

Total harvests for the crop	
1	*harvests*

A5 (Linwood, Gomes, Heifer) - 08/05/2022: Alfalfa, hay

Field name: A5 (Linwood, Gomes, Heifer) Crop: Alfalfa, hay Plant date: 08/05/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	141.41	14.44	123.23	1,304.22
Fresh water	12.45	0.00	0.00	2,287.66
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	167.86	14.44	123.23	3,591.88
Anticipated crop nutrient removal	480.00	43.20	336.00	800.00
Actual crop nutrient removal	495.70	29.24	223.72	731.12
Nutrient balance	-327.84	-14.80	-100.49	2,860.76
Applied to removed ratio	0.34	0.49	0.55	4.91

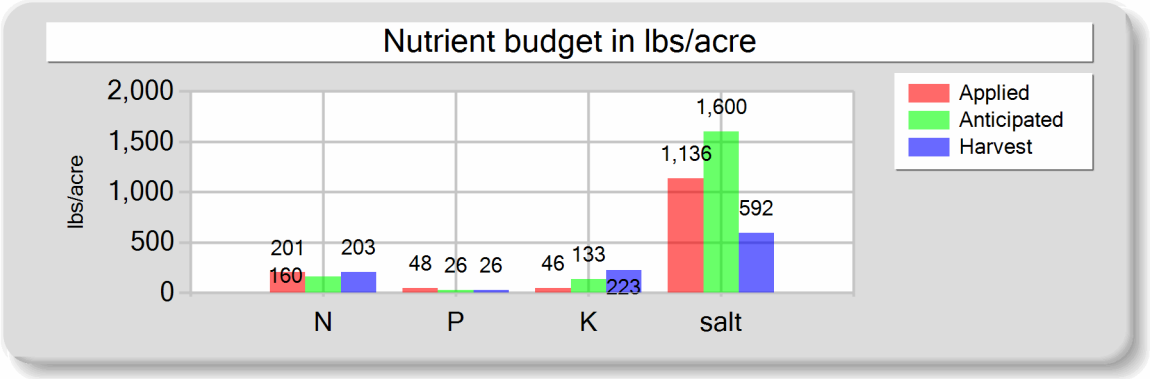
Fresh water applied	
69,000,000.00	*gallons*
2,541.04	*acre-inches*
34.34	*inches/acre*

Process wastewater applied	
4,122,000.00	*gallons*
151.80	*acre-inches*
2.05	*inches/acre*

Total harvests for the crop	
1	*harvests*

A6 (Brown) - 11/14/2022: Oats, silage-soft dough

Field name: A6 (Brown) Crop: Oats, silage-soft dough Plant date: 11/14/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	190.18	47.82	45.86	399.14
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	4.01	0.00	0.00	736.40
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	201.19	47.82	45.86	1,135.54
Anticipated crop nutrient removal	160.00	25.60	132.80	1,600.00
Actual crop nutrient removal	203.22	25.77	222.55	591.51
Nutrient balance	-2.04	22.05	-176.69	544.03
Applied to removed ratio	0.99	1.86	0.21	1.92

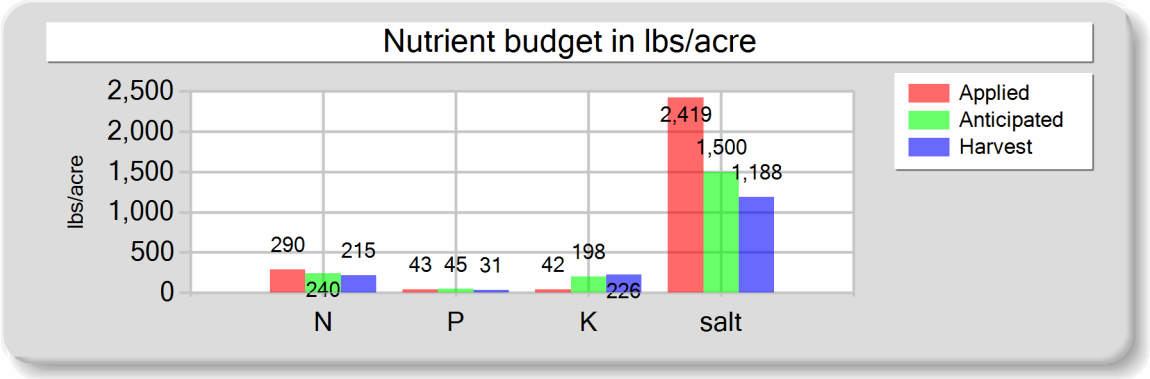
Fresh water applied	
18,009,000.00	*gallons*
663.21	*acre-inches*
11.05	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

A6 (Brown) - 05/17/2023: Corn, silage

Field name: A6 (Brown) Crop: Corn, silage Plant date: 05/17/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	100.00	0.00	0.00	0.00
Dry manure	172.10	43.27	41.50	361.19
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	11.20	0.00	0.00	2,057.36
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	290.29	43.27	41.50	2,418.55
Anticipated crop nutrient removal	240.00	45.00	198.00	1,500.00
Actual crop nutrient removal	215.49	30.75	225.95	1,188.34
Nutrient balance	74.80	12.52	-184.45	1,230.22
Applied to removed ratio	1.35	1.41	0.18	2.04

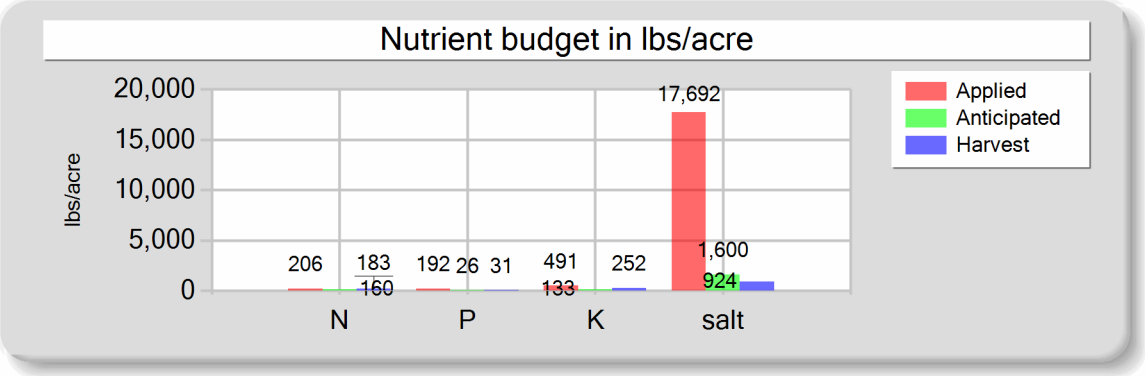
Fresh water applied	
50,314,000.00	*gallons*
1,852.89	*acre-inches*
30.88	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

A8 (Joe Borba-Moffett) - 11/17/2022: Oats, silage-soft dough

Field name: A8 (Joe Borba-Moffett) Crop: Oats, silage-soft dough Plant date: 11/17/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	195.49	192.10	490.53	17,011.47
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	3.70	0.00	0.00	680.40
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	206.19	192.10	490.53	17,691.87
Anticipated crop nutrient removal	160.00	25.60	132.80	1,600.00
Actual crop nutrient removal	182.99	30.73	252.04	923.59
Nutrient balance	23.20	161.38	238.49	16,768.28
Applied to removed ratio	1.13	6.25	1.95	19.16

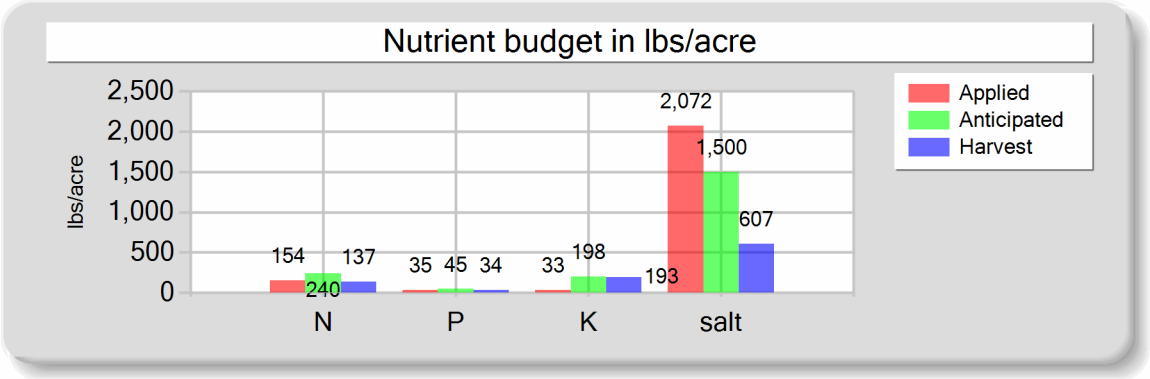
Fresh water applied	
25,513,951.00	*gallons*
939.59	*acre-inches*
10.21	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

A8 (Joe Borba-Moffett) - 06/14/2023: Corn, silage

Field name: A8 (Joe Borba-Moffett) Crop: Corn, silage Plant date: 06/14/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	137.56	34.59	33.17	288.71
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	9.71	0.00	0.00	1,783.38
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	154.27	34.59	33.17	2,072.09
Anticipated crop nutrient removal	240.00	45.00	198.00	1,500.00
Actual crop nutrient removal	136.62	33.85	192.79	607.21
Nutrient balance	17.65	0.74	-159.62	1,464.88
Applied to removed ratio	1.13	1.02	0.17	3.41

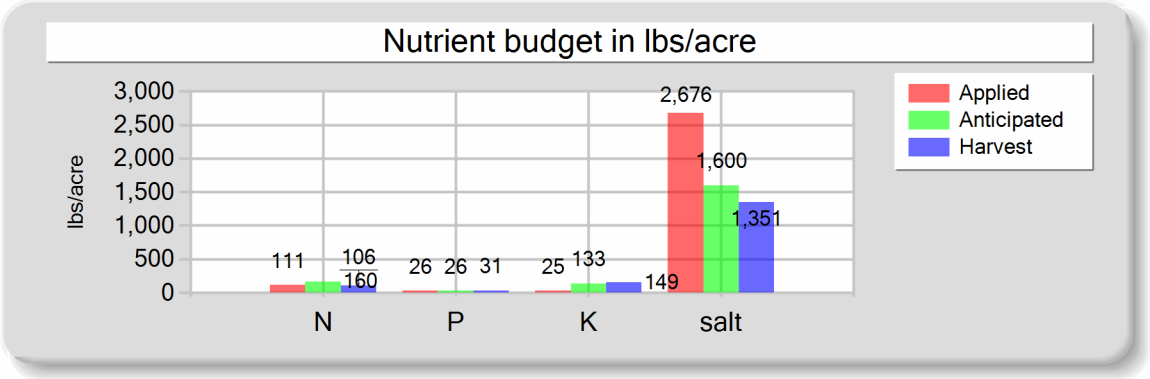
Fresh water applied	
66,874,000.00	*gallons*
2,462.74	*acre-inches*
26.77	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

B1-2 (August) - 10/30/2022: Oats, silage-soft dough

Field name: B1-2 (August) Crop: Oats, silage-soft dough Plant date: 10/30/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	103.67	26.07	25.00	217.57
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	2,458.80
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	110.67	26.07	25.00	2,676.38
Anticipated crop nutrient removal	160.00	25.60	132.80	1,600.00
Actual crop nutrient removal	106.25	31.38	149.17	1,351.01
Nutrient balance	4.41	-5.32	-124.17	1,325.37
Applied to removed ratio	1.04	0.83	0.17	1.98

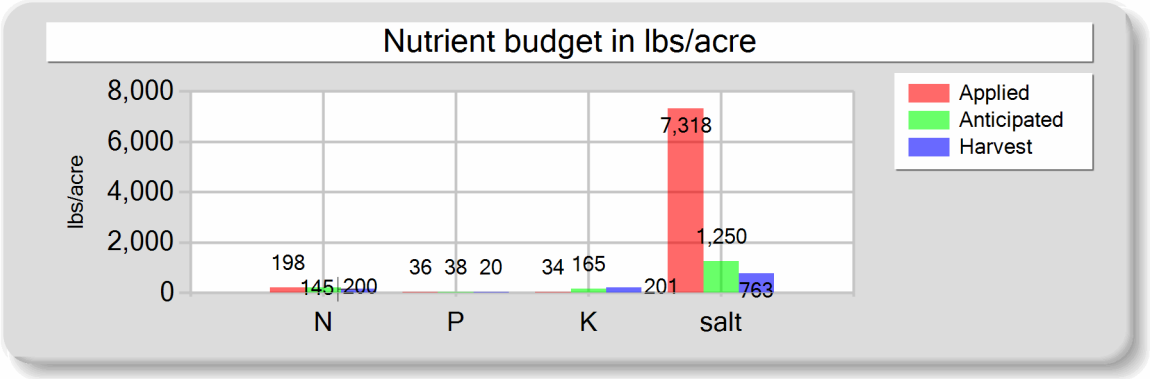
Fresh water applied	
35,278,268.00	*gallons*
1,299.18	*acre-inches*
10.39	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

B1-2 (August) - 06/08/2023: Corn, silage

Field name: B1-2 (August) Crop: Corn, silage Plant date: 06/08/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	50.00	0.00	0.00	0.00
Dry manure	141.38	35.55	34.09	296.71
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	7,021.20
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	198.38	35.55	34.09	7,317.91
Anticipated crop nutrient removal	200.00	37.50	165.00	1,250.00
Actual crop nutrient removal	145.25	19.82	200.91	762.90
Nutrient balance	53.13	15.73	-166.81	6,555.01
Applied to removed ratio	1.37	1.79	0.17	9.59

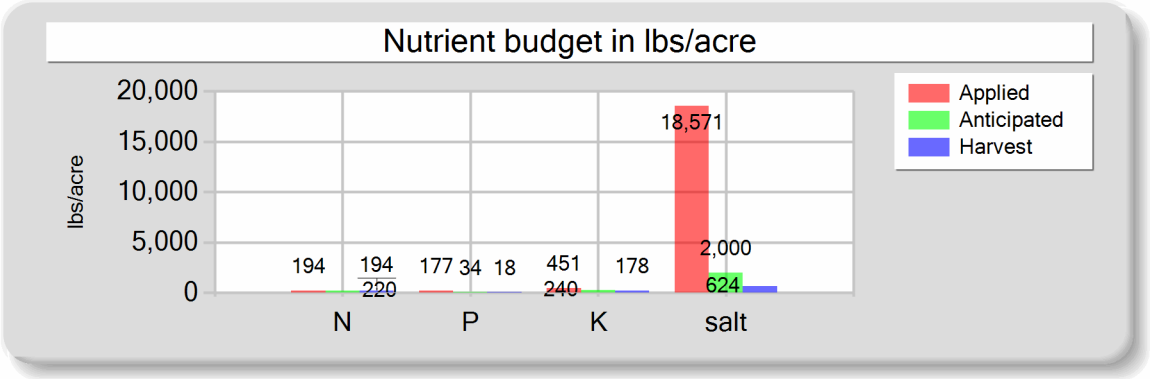
Fresh water applied	
100,738,268.00	*gallons*
3,709.85	*acre-inches*
29.68	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

B3-4 (King) - 07/20/2023: Sudangrass, silage

Field name: B3-4 (King) Crop: Sudangrass, silage Plant date: 07/20/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	179.86	176.75	451.33	15,651.90
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	2,918.65
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	193.86	176.75	451.33	18,570.55
Anticipated crop nutrient removal	220.00	34.00	240.00	2,000.00
Actual crop nutrient removal	193.92	18.49	178.31	624.38
Nutrient balance	-0.05	158.26	273.02	17,946.17
Applied to removed ratio	1.00	9.56	2.53	29.74

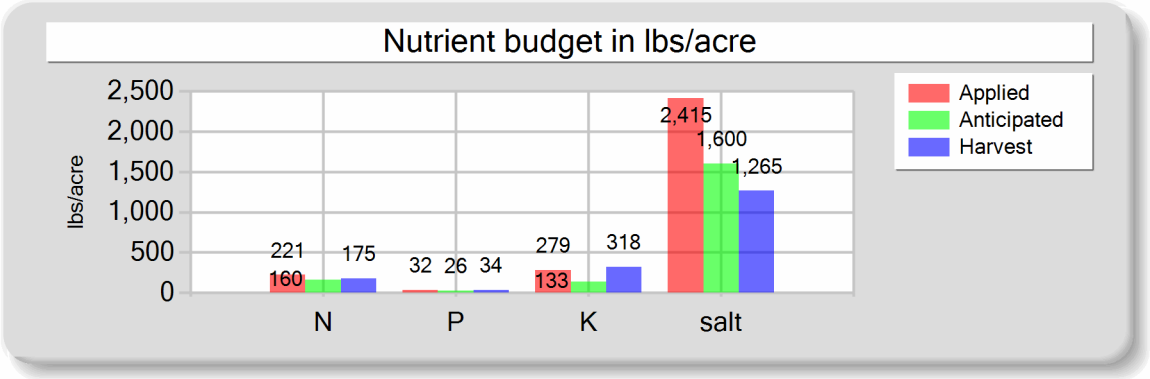
Fresh water applied	
53,601,268.00	*gallons*
1,973.95	*acre-inches*
12.34	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Borba Central - 10/25/2022: Oats, silage-soft dough

Field name: Borba Central | Crop: Oats, silage-soft dough | Plant date: 10/25/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	210.14	32.18	278.59	1,777.19
Fresh water	3.47	0.00	0.00	637.38
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	220.61	32.18	278.59	2,414.56
Anticipated crop nutrient removal	160.00	25.60	132.80	1,600.00
Actual crop nutrient removal	175.46	33.84	317.88	1,264.68
Nutrient balance	45.15	-1.66	-39.29	1,149.89
Applied to removed ratio	1.26	0.95	0.88	1.91

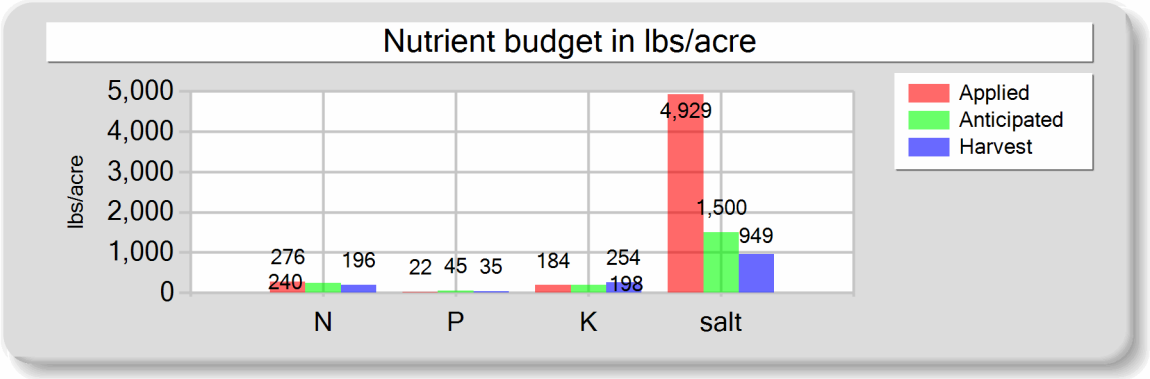
Fresh water applied	
9,872,000.00	*gallons*
363.55	*acre-inches*
9.57	*inches/acre*

Process wastewater applied	
2,734,000.00	*gallons*
100.68	*acre-inches*
2.65	*inches/acre*

Total harvests for the crop	
1	*harvests*

Borba Central - 05/20/2023: Corn, silage

Field name: Borba Central Crop: Corn, silage Plant date: 05/20/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	255.21	21.80	184.42	2,417.92
Fresh water	13.67	0.00	0.00	2,511.29
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	275.88	21.80	184.42	4,929.22
Anticipated crop nutrient removal	240.00	45.00	198.00	1,500.00
Actual crop nutrient removal	196.46	35.12	253.84	949.28
Nutrient balance	79.41	-13.32	-69.42	3,979.93
Applied to removed ratio	1.40	0.62	0.73	5.19

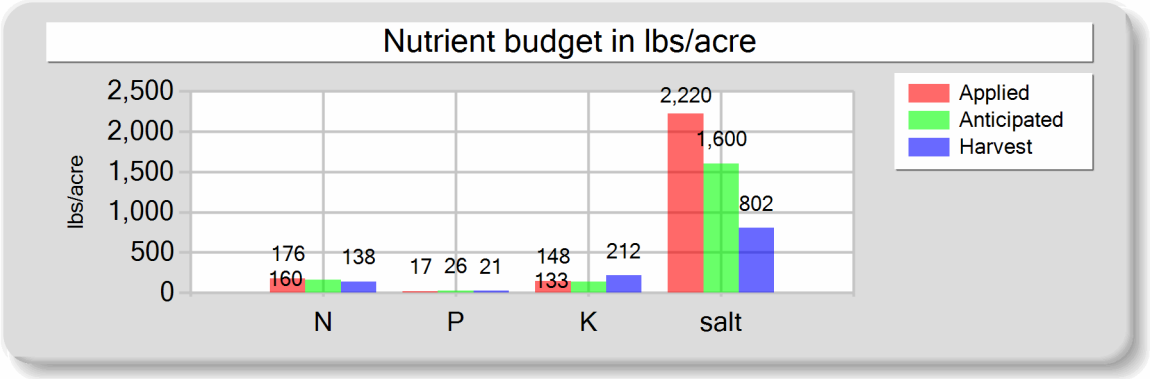
Fresh water applied	
38,896,240.00	*gallons*
1,432.42	*acre-inches*
37.70	*inches/acre*

Process wastewater applied	
3,984,026.00	*gallons*
146.72	*acre-inches*
3.86	*inches/acre*

Total harvests for the crop	
1	*harvests*

Borba-Blaker - 10/22/2022: Oats, silage-soft dough

Field name: Borba-Blaker Crop: Oats, silage-soft dough Plant date: 10/22/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	164.93	17.28	147.64	1,514.61
Fresh water	3.84	0.00	0.00	705.62
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	175.77	17.28	147.64	2,220.23
Anticipated crop nutrient removal	160.00	25.60	132.80	1,600.00
Actual crop nutrient removal	138.41	21.33	211.78	801.73
Nutrient balance	37.36	-4.05	-64.13	1,418.51
Applied to removed ratio	1.27	0.81	0.70	2.77

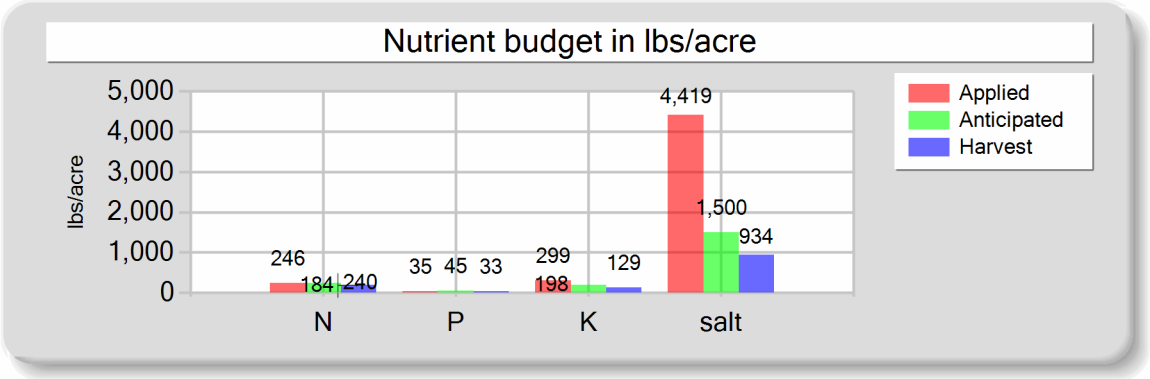
Fresh water applied	
10,929,080.00	*gallons*
402.48	*acre-inches*
10.59	*inches/acre*

Process wastewater applied	
2,452,000.00	*gallons*
90.30	*acre-inches*
2.38	*inches/acre*

Total harvests for the crop	
1	*harvests*

Borba-Blaker - 05/16/2023: Corn, silage

Field name: Borba-Blaker Crop: Corn, silage Plant date: 05/16/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	225.67	34.56	299.17	1,908.49
Fresh water	13.66	0.00	0.00	2,510.84
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	246.33	34.56	299.17	4,419.34
Anticipated crop nutrient removal	240.00	45.00	198.00	1,500.00
Actual crop nutrient removal	184.42	32.94	128.85	934.42
Nutrient balance	61.91	1.62	170.32	3,484.92
Applied to removed ratio	1.34	1.05	2.32	4.73

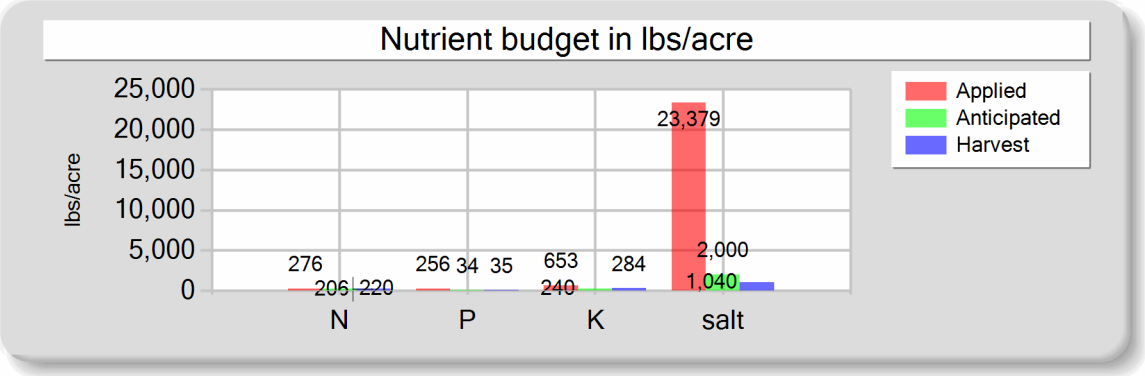
Fresh water applied	
38,889,240.00	*gallons*
1,432.16	*acre-inches*
37.69	*inches/acre*

Process wastewater applied	
2,936,000.00	*gallons*
108.12	*acre-inches*
2.85	*inches/acre*

Total harvests for the crop	
1	*harvests*

R1-2 (River) - 07/17/2023: Sudangrass, silage

Field name: R1-2 (River) Crop: Sudangrass, silage Plant date: 07/17/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	260.30	255.80	653.17	22,651.83
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	1.48	0.00	0.00	726.99
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	275.79	255.80	653.17	23,378.82
Anticipated crop nutrient removal	220.00	34.00	240.00	2,000.00
Actual crop nutrient removal	206.02	34.60	283.76	1,039.80
Nutrient balance	69.77	221.20	369.41	22,339.02
Applied to removed ratio	1.34	7.39	2.30	22.48

Fresh water applied	
37,070,853.00	*gallons*
1,365.19	*acre-inches*
13.65	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

T3277014A

Sample and source description: T3277014A

Sample date: 10/04/2023 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 81.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	24,300.00	6,110.00	5,860.00							5.10
DL	100.00	100.00	100.00							0.10

T4002005

Sample and source description: T4002005

Sample date: 01/02/2024 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 23.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	8,090.00	7,950.00	20,300.00	8,430.00	3,440.00	2,910.00	2,310.00	648.00		70.40
DL	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00		0.10

B. PROCESS WASTEWATER ANALYSES

T3184016

Sample and source description: T3184016

Sample date: 03/15/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	666.00	462.00	0.00	0.00	33.30	735.00								8,970.00	5,740
DL	0.10	0.10	0.10	0.10	0.10	0.10								20.00	10

T3130014B

Sample and source description: T3130014B

Sample date: 05/09/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	350.00	176.00	0.00	0.00	53.60	464.00								5,180.00	2,960
DL	0.10	0.10	0.10	0.10	0.10	0.10								20.00	10

T3237007A

Sample and source description: T3237007A

Sample date: 08/25/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	255.00	165.00	0.00	0.00	6.86	51.40								4,030.00	2,640
DL	0.10	0.10	0.10	0.10	0.10	0.10								20.00	10

T3352013A

Sample and source description: T3352013A

Sample date: 12/18/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	420.00	288.00	0.00	0.00	48.10	503.00								5,860.00	4,700
DL	0.10	0.10	0.10	0.10	0.10	0.10								20.00	10

C. FRESH WATER ANALYSES

Ag Well 3

Ag Well 3

T3345002C

Sample description: T3345002C

Sample date: 12/09/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								1,740.00	1,044
DL	0.10	0.10	0.10								20.00	10

San Joaquin River

CEDEN data

Sample description: CEDEN data

Sample date: 06/12/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.48	0.00	0.48								391.00	235
DL	0.10	0.10	0.10								20.00	10

Turlock Irrigation District

TID info

Sample description: TID info

Sample date: 06/20/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1.60	0.00	1.60								490.00	294
DL	0.10	0.10	0.10								20.00	10

D. SOIL ANALYSES

No soil analyses entered.

E. PLANT TISSUE ANALYSES

A1-Dairy Field - 11/11/2022: Oats, silage-soft dough

T3117023A

Sample and source description: T3117023A

Sample date: 04/26/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 66.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	32,300.00	3,080.00	29,700.00		10.40
DL	100.00	100.00	100.00		0.10

A1-Dairy Field - 06/10/2023: Corn, silage

T3142015A

Sample and source description: T3142015A

Sample date: 09/15/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 70.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,000.00	1,970.00	15,900.00		6.57
DL	100.00	100.00	100.00		0.10

A3-4 (Silva, Lima) - 11/13/2022: Oats, silage-soft dough

A3-4 (Silva, Lima) - 11/13/2022: Oats, silage-soft dough

T3111006A

Sample and source description: T3111006A

Sample date: 04/20/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 77.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	32,500.00	5,000.00	59,000.00		12.50
DL	100.00	100.00	100.00		0.10

A3-4 (Silva, Lima) - 05/14/2023: Corn, silage

T3254023B

Sample and source description: T3254023B

Sample date: 09/08/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 67.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,100.00	1,950.00	14,200.00		5.30
DL	100.00	100.00	100.00		0.10

A5 (Linwood, Gomes, Heifer) - 08/05/2022: Alfalfa, hay

774850A

Sample and source description: 774850A

Sample date: 07/19/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 8.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	33,900.00	2,000.00	15,300.00		5.00
DL	100.00	100.00	100.00		0.10

A6 (Brown) - 11/14/2022: Oats, silage-soft dough

T3111006B

Sample and source description: T3111006B

Sample date: 05/17/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 68.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	34,700.00	4,400.00	38,000.00		10.10
DL	100.00	100.00	100.00		0.10

A6 (Brown) - 05/17/2023: Corn, silage

T3262011B

Sample and source description: T3262011B

Sample date: 09/18/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 64.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	10,300.00	1,470.00	10,800.00		5.68
DL	100.00	100.00	100.00		0.10

A8 (Joe Borba-Moffett) - 11/17/2022: Oats, silage-soft dough

T3130014A

Sample and source description: T3130014A

Sample date: 05/08/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 71.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	21,200.00	3,560.00	29,200.00		10.70
DL	100.00	100.00	100.00		0.10

A8 (Joe Borba-Moffett) - 06/14/2023: Corn, silage

T3275022A

Sample and source description: T3275022A

Sample date: 09/29/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 74.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	9,000.00	2,230.00	12,700.00		4.00
DL	100.00	100.00	100.00		0.10

B1-2 (August) - 10/30/2022: Oats, silage-soft dough

T3142015A

Sample and source description: T3142015A

Sample date: 05/22/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 55.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	7,550.00	2,230.00	10,600.00		9.60
DL	100.00	100.00	100.00		0.10

B1-2 (August) - 06/08/2023: Corn, silage

T3326015B

Sample and source description: T3326015B

Sample date: 10/04/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 70.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	10,700.00	1,460.00	14,800.00		5.62
DL	100.00	100.00	100.00		0.10

B3-4 (King) - 07/20/2023: Sudangrass, silage

T3117023

Sample and source description: T3117023

Sample date: 10/17/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 66.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	32,300.00	3,080.00	29,700.00		10.40
DL	100.00	100.00	100.00		0.10

Borba Central - 10/25/2022: Oats, silage-soft dough

T3144038

Sample and source description: T3144038

Sample date: 04/12/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 63.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,400.00	2,970.00	27,900.00		11.10
DL	100.00	100.00	100.00		0.10

Borba Central - 05/20/2023: Corn, silage

T3254023A

Sample and source description: T3254023A

Sample date: 09/08/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 71.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,300.00	2,020.00	14,600.00		5.46
DL	100.00	100.00	100.00		0.10

Borba-Blaker - 10/22/2022: Oats, silage-soft dough

T3144038A

Sample and source description: T3144038A

Sample date: 05/24/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 74.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	18,300.00	2,820.00	28,000.00		10.60
DL	100.00	100.00	100.00		0.10

Borba-Blaker - 05/16/2023: Corn, silage

T3326015A

Sample and source description: T3326015A

Sample date: 10/12/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 56.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	8,230.00	1,470.00	5,750.00		4.17
DL	100.00	100.00	100.00		0.10

R1-2 (River) - 07/17/2023: Sudangrass, silage

T3130014

Sample and source description: T3130014

Sample date: 10/18/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 71.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	21,200.00	3,560.00	29,200.00		10.70
DL	100.00	100.00	100.00		0.10

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

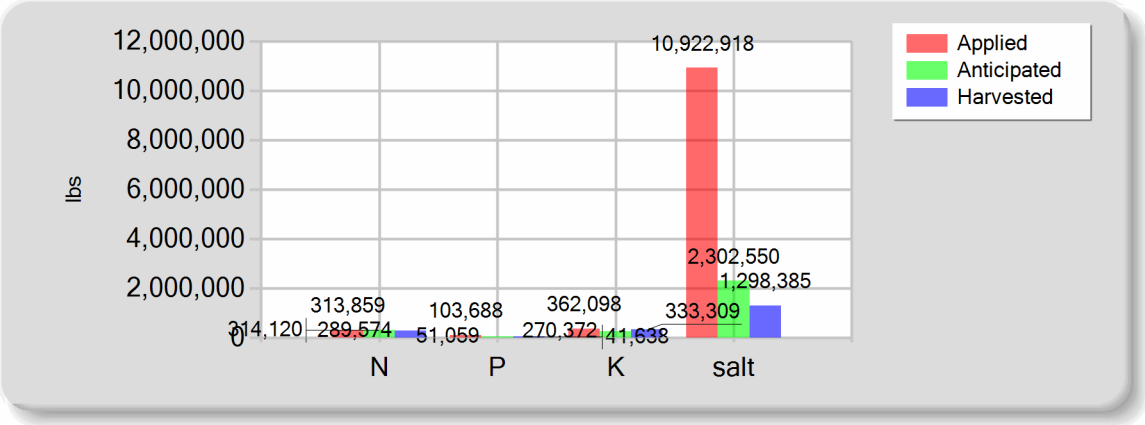
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

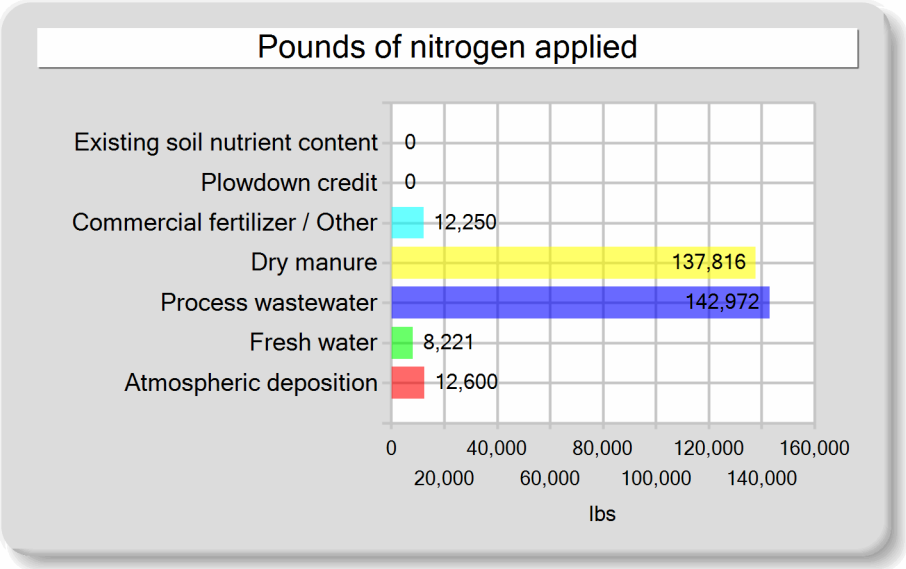
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

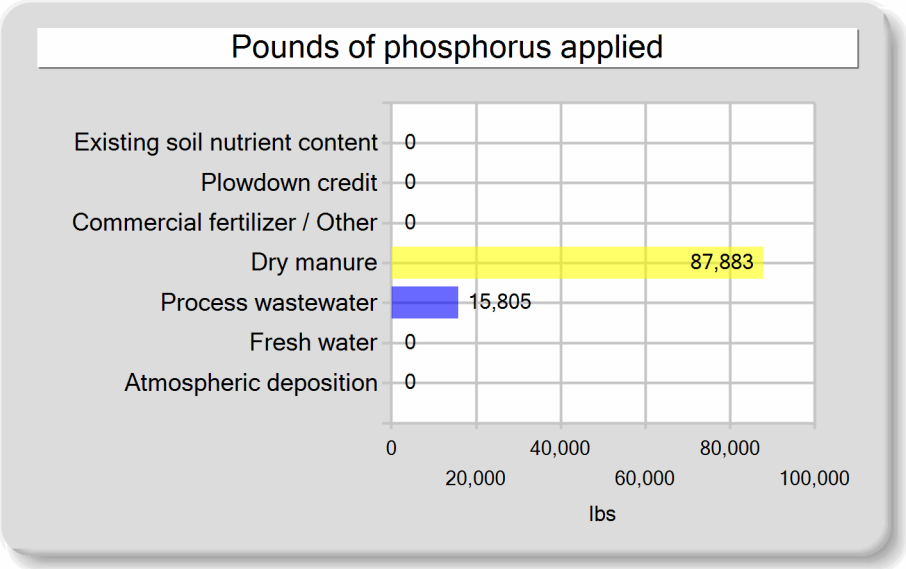
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	12,250.00	0.00	0.00	0.00
Dry manure	137,815.83	87,882.84	198,338.32	6,471,009.85
Process wastewater	142,972.02	15,804.76	163,760.03	1,243,903.67
Fresh water	8,220.99	0.00	0.00	3,208,004.32
Atmospheric deposition	12,600.00	0.00	0.00	0.00
Total nutrients applied	313,858.83	103,687.60	362,098.35	10,922,917.84
Anticipated crop nutrient removal	314,120.00	51,058.90	270,371.80	2,302,550.00
Actual crop nutrient removal	289,573.64	41,637.99	333,308.94	1,298,385.11
Nutrient balance	24,285.19	62,049.60	28,789.41	9,624,532.72
Applied to removed ratio	1.08	2.49	1.09	8.41

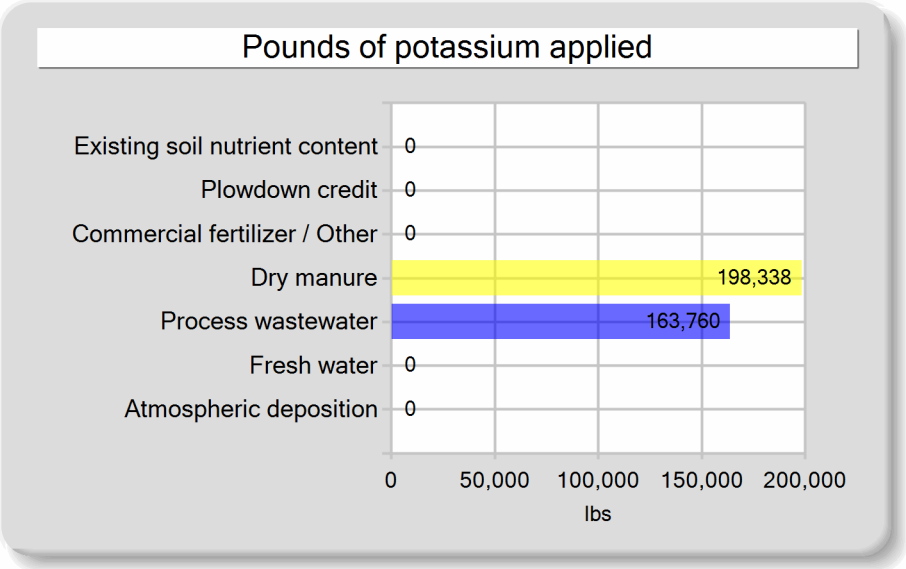
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

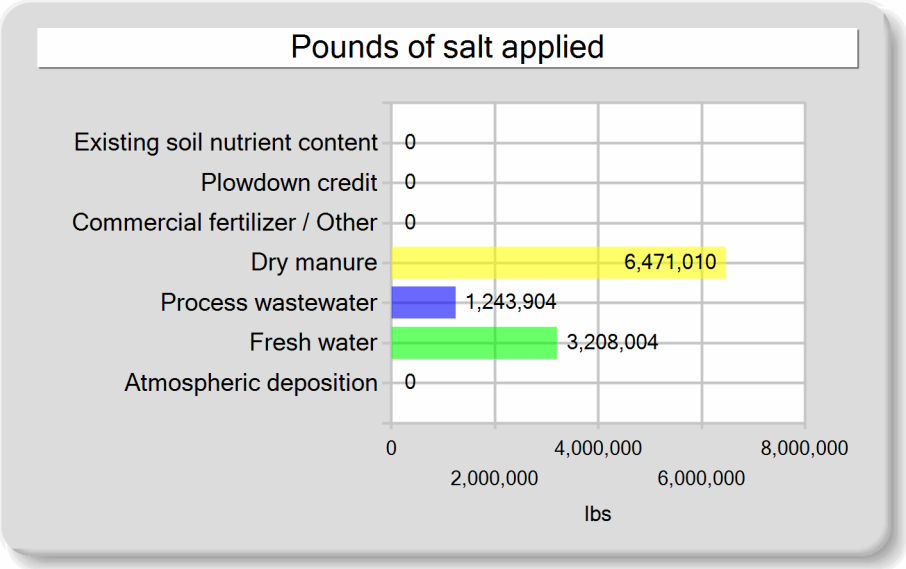


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	No
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

No notes entered for this annual report.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

Armelim de Sousa

SIGNATURE OF OWNER OF FACILITY

SIGNATURE OF OPERATOR OF FACILITY

Armelim Sousa

PRINT OR TYPE NAME

SAME AS OWNER

PRINT OR TYPE NAME

6-30-24

DATE

DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

Production Well Water Quality Analysis 2023

A & M de Sousa Dairy, 3030 S. Central Ave., Turlock

Well Name	Location	Date Sampled	EC	Ammonia	Nitrate-N (mg/L)
Well #1	Dairy 5hp	10/15/2023	1110	ND	35.3
Well #2	Dairy 10hp	10/15/2023	1150	ND	30.3
Well #3	Log Cabin	10/15/2023	1500	ND	55.7
Well #4	Tony's	10/15/2023	1900	ND	56.6
Well T1	Thomas Heifer Ranch	10/15/2023	1210	ND	20.9
Well #10	August Houses	12/9/2023	2080	ND	ND
Ag Well #3	August 75hp	12/9/2023	1740	ND	ND
Ag Well #4	August bottom well	12/9/2023	2040	ND	ND
Ag Well #5	Dairy 75hp	10/15/2023	1225	ND	25.1
Ag Well #6	Turkey east	10/15/2023	1290	ND	53.7
Ag Well #7	Turkey west	10/15/2023	1800	ND	64.4

ND = Non Detect

Denele Analytical, Inc.

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40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

www.denelelabs.com

Compliance Analysis Report

Customer
A & M De Sousa Dairy
3030 S. Central Ave.
Turlock, CA 95380

Grower

Consultant:
Paul Sousa

PURCHASE ORDER:
RECEIVED DATE: 10/16/23 3:24 PM
SUBMITTED BY: Paul Sousa
ANALYZED DATE: 10/17/23

Nitrate Nitrogen (NO3-N)
SOURCE: Well Water

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Well 1	T3290009A	Nitrate Nitrogen (NO3-N)	EPA 353.2	35.3	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Nitrate Nitrogen (NO3-N)	98.8	101 (W3289001A)	103 (W3289001A)	1.35

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

ELAP Certificate No.: 2714

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Grower

Consultant:
Paul Sousa

PURCHASE ORDER:
RECEIVED DATE: 10/16/23 3:24 PM
SUBMITTED BY: Paul Sousa
ANALYZED DATE: 10/17/23

Nitrate Nitrogen (NO3-N)
SOURCE: Well Water

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Well 2	T3290009B	Nitrate Nitrogen (NO3-N)	EPA 353.2	30.3	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Nitrate Nitrogen (NO3-N)	98.8	101 (W3289001A)	103 (W3289001A)	1.35

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Customer
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Turlock, CA 95380

Grower

Consultant:
Paul Sousa

PURCHASE ORDER:
RECEIVED DATE: 10/16/23 3:24 PM
SUBMITTED BY: Paul Sousa
ANALYZED DATE: 10/17/23

Nitrate Nitrogen (NO3-N)
SOURCE: Well Water

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Well 3	T3290009C	Nitrate Nitrogen (NO3-N)	EPA 353.2	55.7	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Nitrate Nitrogen (NO3-N)	98.8	101 (W3289001A)	103 (W3289001A)	1.35

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Customer
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Grower

Consultant:
Paul Sousa

PURCHASE ORDER:
RECEIVED DATE: 10/16/23 3:24 PM
SUBMITTED BY: Paul Sousa
ANALYZED DATE: 10/17/23

Nitrate Nitrogen (NO3-N)
SOURCE: Well Water

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Well 4	T3290009D	Nitrate Nitrogen (NO3-N)	EPA 353.2	56.6	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Nitrate Nitrogen (NO3-N)	98.8	101 (W3289001A)	103 (W3289001A)	1.35

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Compliance Analysis Report

Customer
A & M De Sousa Dairy
3030 S. Central Ave.
Turlock, CA 95380

Grower

Consultant:
Paul Sousa

PURCHASE ORDER:
RECEIVED DATE: 10/16/23 3:24 PM
SUBMITTED BY: Paul Sousa
ANALYZED DATE: 10/17/23

Nitrate Nitrogen (NO3-N)
SOURCE: Well Water

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
T1	T3290009E	Nitrate Nitrogen (NO3-N)	EPA 353.2	20.9	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Nitrate Nitrogen (NO3-N)	98.8	101 (W3289001A)	103 (W3289001A)	1.35

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Compliance Analysis Report

Customer
A & M De Sousa Dairy
3030 S. Central Ave.
Turlock, CA 95380

Grower

Consultant:
Paul Sousa

PURCHASE ORDER:
RECEIVED DATE: 10/16/23 3:24 PM
SUBMITTED BY: Paul Sousa
ANALYZED DATE: 10/17/23

Nitrate Nitrogen (NO3-N)
SOURCE: Well Water

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
AG 5	T3290009F	Nitrate Nitrogen (NO3-N)	EPA 353.2	25.1	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Nitrate Nitrogen (NO3-N)	98.8	101 (W3289001A)	103 (W3289001A)	1.35

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Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

www.denelelabs.com

Compliance Analysis Report

Customer
A & M De Sousa Dairy
3030 S. Central Ave.
Turlock, CA 95380

Grower

Consultant:
Paul Sousa

PURCHASE ORDER:
RECEIVED DATE: 10/16/23 3:24 PM
SUBMITTED BY: Paul Sousa
ANALYZED DATE: 10/17/23

Nitrate Nitrogen (NO3-N)
SOURCE: Well Water

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
AG 6	T3290009G	Nitrate Nitrogen (NO3-N)	EPA 353.2	53.7	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Nitrate Nitrogen (NO3-N)	98.8	101 (W3289001A)	103 (W3289001A)	1.35

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Compliance Analysis Report

Customer
A & M De Sousa Dairy
3030 S. Central Ave.
Turlock, CA 95380

Grower

Consultant:
Paul Sousa

PURCHASE ORDER:
RECEIVED DATE: 10/16/23 3:24 PM
SUBMITTED BY: Paul Sousa
ANALYZED DATE: 10/17/23

Nitrate Nitrogen (NO3-N)
SOURCE: Well Water

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
AG 7	T3290009H	Nitrate Nitrogen (NO3-N)	EPA 353.2	64.4	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Nitrate Nitrogen (NO3-N)	98.8	101 (W3289001A)	103 (W3289001A)	1.35

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DENELE ANALYTICAL, INC.

865 S. KILROY
TURLOCK, CA 95380

PH# 209-634-9055 FAX# 209-634-9057

DAIRY COMPLIANCE

Client: A&M de Sousa
Mailing address: 3030 S Central Ave
(not facility address) Turlock 95380
Phone#:

Authorized Copy Released To: Paul Sousa

Sampled By: Paul Sousa
Project ID:

ANALYSIS TO BE COMPLETED:

IRRIGATION/GROUND WATER
- H1 EC*, NO3-N, NH4-N*
- H6 EC, TN, TDS
- H7 Combination of H1 & H6
- H14 EC, NO3-N, NH4-N, Ca, Mg, Na, Bicarb Carb, SO4, Cl, TDS

PROCESS WASTE WATER
- H3 EC*, NH4-N, TKN, TP, K, TDS
- H4 EC*, NO3-N, NH4-N, TKN, TP, K, TDS
- H5 Ca, Mg, Na, Bicarb, Carb, SO4, Cl
- H8 Combination of H3 &H5
- H9 Combination of H4 &H5

TILE DRAIN
- H2 EC*, NO3-N, TP, NH4-N*, TDS

** Required Field Measurement*

___ (Check here if performed)

DISCHARGE WATER
- E1 EC*, Temp*, pH*, NO3-N, NH4-N, TKN, TP, K, TDS, Dissolved Oxygen*, BOD 5, TSS, Total & Fecal Coliforms
- E2 EC,* Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Dissolved Oxygen*, Total & Fecal Coliforms
- E3 EC*, Temp*, pH*, NO3-N, NH4-N*, TP, Total & Fecal Coliforms
- E4 EC*, Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Total & Fecal Coliform

MANURE
- M1 % Moisture
- M3 TN TP, K, % Moisture
- M4 Ca, Mg, Na, Bicarb, Carb, SO4, Cl, Ash
- M7 Combination of M3 & M4

PLANT TISSUE
- F18 % Moisture, Ash, TN, P, K, DM and as received
- F18+ % Moisture, Ash, TN, P, K, DM and as received (Package also includes NIR DCAD)
- F19+ % Moisture, TN (Package includes complete plant tissue analysis)

SOIL
- K3 Soluble Phosphorus
- K4 1ft NO3, Organic Matter
- K5 Pre-Plant (2ft NO3), Fall (1-2ft NO3), Fall (2-3ft NO3)
- K6 0-1ft EC, NO3, Soluble P, K, Organic Matter
- K7 Combination of K3 & K4
- K8 Combination of K3 & K5

	Date Collected	Time Collected	Sample ID	Description	Temp	Well Purge	Analysis: H1	H3	M3	F18	OTHER
1	10/15/23	4:15	Well 1		1.8°c		NO_3-N				
2		4:25	Well 2		2.8°c						
3		4:40	Well 3		1.4°c						
4		4:30	Well 4		2.0°c						
5		5:07	T1		1.6°c						
6		5:25	AG5		1.8°c						
7		5:40	AG6		2.6°c						
8	10/15/23	5:20	AG7		4.8°c		NO_3-N				
9											
10											

CHAIN OF CUSTODY RECORDING

	Signature	Company	Received Date & Time	Relinquished Date & Time	(LABORATORY USE ONLY) Logged in By:	LABORATORY NUMBER
1st	Paul Sousa					
2nd		DENELE ANALYTICAL, INC.	10-16-23 @ 15:24			T3290009

Denele Analytical, Inc.

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www.denelelabs.com

Compliance Analysis Report

Customer
A & M De Sousa Dairy
3030 S. Central Ave.
Turlock, CA 95380

Grower

Consultant:
Paul Sousa

PURCHASE ORDER:
RECEIVED DATE: 12/11/23 9:22 AM
SUBMITTED BY: Paul
ANALYZED DATE: 12/11/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Well 10	T3345002B	Electrical Conductivity (EC)	EPA 120.1 and	2.08	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	< 0.140	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	1330	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	90.4	98.2 (T3342032A)	100 (T3342032A)	1.81
Electrical Conductivity (EC)	95.9			2.19
Nitrate Nitrogen (NO3-N)	100	105 (T3347004A)	102 (T3347004A)	3.32

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ELAP Certificate No.: 2714

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Compliance Analysis Report

Customer
A & M De Sousa Dairy
3030 S. Central Ave.
Turlock, CA 95380

Grower

Consultant:
Paul Sousa

PURCHASE ORDER:
RECEIVED DATE: 12/11/23 9:22 AM
SUBMITTED BY: Paul
ANALYZED DATE: 12/11/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Well AG 3	T3345002C	Electrical Conductivity (EC)	EPA 120.1 and	1.74	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	< 0.140	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	1110	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	90.4	98.2 (T3342032A)	100 (T3342032A)	1.81
Electrical Conductivity (EC)	95.9			2.19
Nitrate Nitrogen (NO3-N)	100	105 (T3347004A)	102 (T3347004A)	3.32

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Compliance Analysis Report

Customer
A & M De Sousa Dairy
3030 S. Central Ave.
Turlock, CA 95380

Grower

Consultant:
Paul Sousa

PURCHASE ORDER:
RECEIVED DATE: 12/11/23 9:22 AM
SUBMITTED BY: Paul
ANALYZED DATE: 12/11/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Well AG 4	T3345002D	Electrical Conductivity (EC)	EPA 120.1 and	2.04	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	< 0.140	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	1300	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	90.4	98.2 (T3342032A)	100 (T3342032A)	1.81
Electrical Conductivity (EC)	95.9			2.19
Nitrate Nitrogen (NO3-N)	100	105 (T3347004A)	102 (T3347004A)	3.32

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DENELE ANALYTICAL, INC.

865 S. KILROY
TURLOCK, CA 95380

PH# 209-634-9055 FAX# 209-634-9057

DAIRY COMPLIANCE

Client: A&M de Sousa Dairy
Mailing address: 3030 S. Central Ave
(not facility address) Turlock 95380
Phone#:

Authorized Copy Released To: Paul Sousa

Sampled By: Paul
Project ID:

ANALYSIS TO BE COMPLETED:

IRRIGATION/GROUND WATER
- H1 EC*, NO3-N, NH4-N*
- H6 EC, TN, TDS
- H7 Combination of H1 & H6
- H14 EC, NO3-N, NH4-N, Ca, Mg, Na, Bicarb Carb, SO4, Cl, TDS

PROCESS WASTE WATER
- H3 EC*, NH4-N, TKN, TP, K, TDS
- H4 EC*, NO3-N, NH4-N, TKN, TP, K, TDS
- H5 Ca, Mg, Na, Bicarb, Carb, SO4, Cl
- H8 Combination of H3 &H5
- H9 Combination of H4 &H5

TILE DRAIN
- H2 EC*, NO3-N, TP, NH4-N*, TDS

** Required Field Measurement*

___ (Check here if performed)

DISCHARGE WATER
- E1 EC*, Temp*, pH*, NO3-N, NH4-N, TKN, TP, K, TDS, Dissolved Oxygen*, BOD 5, TSS, Total & Fecal Coliforms
- E2 EC,* Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Dissolved Oxygen*, Total & Fecal Coliforms
- E3 EC*, Temp*, pH*, NO3-N, NH4-N*, TP, Total & Fecal Coliforms
- E4 EC*, Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Total & Fecal Coliform

MANURE
- M1 % Moisture
- M3 TN TP, K, % Moisture
- M4 Ca, Mg, Na, Bicarb, Carb, SO4, Cl, Ash
- M7 Combination of M3 & M4

PLANT TISSUE
- F18 % Moisture, Ash, TN, P, K, DM and as received
- F18+ % Moisture, Ash, TN, P, K, DM and as received (Package also includes NIR DCAD)
- F19+ % Moisture, TN (Package includes complete plant tissue analysis)

SOIL
- K3 Soluble Phosphorus
- K4 1ft NO3, Organic Matter
- K5 Pre-Plant (2ft NO3), Fall (1-2ft NO3), Fall (2-3ft NO3)
- K6 0-1ft EC, NO3, Soluble P, K, Organic Matter
- K7 Combination of K3 & K4
- K8 Combination of K3 & K5

	Date Collected	Time Collected	Sample ID	Description	Temp	Well Purge	Analysis: H1	Analysis: H3	Analysis: M3	Analysis: F18	Analysis: OTHER
1	12/9/23	2:32	Well M1		0.8		NO_3-N				
2		2:42	Well 10		0.6		↓				
3		2:45	Well Ag 3		0.2		↓				
4	12/9/23	2:50	Well Ag 4		1.2		NO_3-N				
5	12/9/23	2:15	Heifer Manure						M7		
6											
7											
8											
9											
10											

CHAIN OF CUSTODY RECORDING

	Signature	Company	Received Date & Time	Relinquished Date & Time	(LABORATORY USE ONLY) Logged in By:	LABORATORY NUMBER
1st	Pal Sousa					
2nd	[illegible]	DENELE ANALYTICAL, INC.	12-11-23 @ 9:22			T3345022

Required to be submitted to any agency? (YES / NO)